

integration

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1 Notes

Let $(X, \mathcal{M}, \mu)$ be a measure space. $f : X \rightarrow \mathbb{R}$ is said to be simple if $\text{im } f$ is finite and f is measurable.

Claim: Let f, g be measurable. Then $\lambda f, f + g$ are measurable, where $\lambda \in \mathbb{R}$.

Proof: Since f, g are measurable, so is $f + g$. Further, $\text{im } (f + g) \leq \text{im } f + \text{im } g$, which is finite, since $\text{im } f \times \text{im } g \rightarrow \text{im } f + \text{im } g$ (given by addition) is surjective, and the domain is finite. λf is measurable and $\text{im } (\lambda f) = \lambda \text{im } f$.

Suppose f is simple. Let $\{X_i\} \leq \mathcal{M}$ be a finite partition of X such that f is constant with value a_i on X_i (if $X_i = \emptyset$, take a_i to be arbitrary). Define $\int_X f d\mu = \sum_i a_i \mu(X_i)$.

Claim: $\int_X f d\mu$ is well-defined.

Proof: Let $\{X'_j\} \leq \mathcal{M}$ be another finite partition of X with f constant over the X'_j with values a'_j . Then $\{X_i \cap X'_j\}$ is a finite partition of X . We have

$$\sum_i a_i \mu(X_i) = \sum_i a_i \sum_j \mu(X_i \cap X'_j) = \sum_i \sum_j a_i \mu(X_i \cap X'_j) = \sum_i \sum_j a'_j \mu(X_i \cap X'_j)$$

The last step is valid since if $X_i \cap X'_j = \emptyset$, then $a_i \mu(X_i \cap X'_j) = a'_j \mu(X_i \cap X'_j)$ for any a_i, a'_j . Else, a_i, a'_j equal to f evaluated at some value in $X_i \cap X'_j$. Fubini's theorem lets us swap the sums.

Suppose f is simple. Then we may write $f = \sum_i a_i 1_{E_i}$, where $E_i = f^{-1}a_i \in \mathcal{M}$. The sets $\{E_i\}$ form a finite partition of X into elements of $\mathcal{M}$. Thus, $\int_X f d\mu = \sum_i a_i \mu(E_i)$.

Claim: If $f = \sum_i a_i 1_{E_i}$, where the E_i need not be disjoint, then f is simple.

Proof: The image of f is contained in the sums of the multisubsets of multiset $\{a_1, a_2, \dots, a_n\}$, so $\text{im } f$ is finite. Since 1_{E_i} is measurable, so is f (as a linear combination of measurable functions).

Claim: If $f = \sum_i a_i 1_{E_i}$, where the E_i need not be disjoint, then $\int_X f d\mu = \sum_i a_i \mu(E_i)$.

Proof: We strengthen the claim to asserting the existence of disjoint F_j contained in $\bigcup_i E_i$ such that $f = \sum_j b_j 1_{F_j}$, and that f is constant on the F_j , and proceed by induction. Denote by n the number of E_i . If $n = 0, 1$, we're done. Assume the result for n , and consider $n + 1$ E_i 's. Let $g_0 = 1_{E_{n+1}} f$ and $g_1 = 1_{E_{n+1}^c} f$. Note that $g_0 + g_1 = f$ and $g_0 = 1_{E_{n+1}} \sum_i a_i 1_{E_i} = \sum_i a_i 1_{E_i \cap E_{n+1}}$, so g_0, g_1 are sums of n indicator functions. Note that g_0 is sum of n indicator functions plus $a_{n+1} 1_{E_{n+1}}$. g_1 is a sum of n indicator functions.

By the IH, write $g_0 = a_{n+1} 1_{E_{n+1}} + \sum_j b_j 1_{F_j}$ and $g_1 = \sum_k c_k 1_{G_k}$ such that the F_j 's are disjoint, f is piecewise constant on the F_j , and contained in E_{n+1} , and the G_k 's are disjoint, f is constant on the G_k , and contained in E_{n+1}^c . Then $f = a_{n+1} 1_{E_{n+1}} + \sum_j b_j 1_{F_j} + \sum_k c_k 1_{G_k}$, giving a presentation of f in the desired form. Then

$$\begin{aligned} \int_X f d\mu &= a_{n+1} \mu(E_{n+1}) + \sum_j b_j \mu(F_j) + \sum_k c_k \mu(G_k) \\ &= a_{n+1} \mu(E_{n+1}) + \sum_{i \in [n]} \mu(E_i \cap E_{n+1}) + \sum_{i \in [n]} a_i \mu(E_i \cap E_{n+1}^c) \\ &= \sum_i a_i \mu(E_i) \end{aligned}$$

*****fix later

Claim: Let s be simple and $\{E_i\}_{i \in \mathbb{Z}^+}$ be disjoint. Then $\int_E s = \sum_{i \in \mathbb{Z}^+} \int_{E_i} s$.

Proof: Write $s = \sum_j a_j 1_{A_j}$. Then

$$\begin{aligned} \int_E s &= \int_X 1_E \cdot s = \int_X \sum_j a_j 1_{E \cap A_j} = \sum_j a_j \mu(E \cap A_j) \\ &= \sum_j a_j \sum_i \mu(E_i \cap A_j) = \sum_i \sum_j a_j \mu(E_i \cap A_j) = \sum_i \int_{E_i} s \end{aligned}$$

as desired.

Claim: Let s be simple and $\{E_i\}$ be ascending and $E = \bigcup_{n \in \mathbb{Z}^+} E_n$. Then $\int_E s = \lim_{n \rightarrow \infty} \int_{E_n} s$.

Proof: The map $A \mapsto \int_A s$ is a measure. The claim is continuity from below.

Claim (Monotone convergence theorem): Let (f_n) be increasing with limit f . Then $\lim \int f_n = \int f$.

Proof: Let $0 \leq \phi \leq f$ be simple and $E_n = \{x \in X : f_n(x) \geq (1 - \alpha)\phi(x), \phi(x) < \infty\}$, so $E_n \in \mathcal{M}$. I claim that $\bigcup_{n \in \mathbb{Z}^+} E_n = E = \phi^{-1}([0, \infty))$. One direction is clear. Suppose $x \in \phi^{-1}([0, \infty))$, so $\phi(x) < \infty$. If $\phi(x) = 0$, we're done, since $x \in E_1$. If $0 < \phi(x) < \infty$, then $(1 - \alpha)\phi(x) < \phi(x) \leq f(x)$. Thus, there exists $n \in \mathbb{Z}^+$ such that $f_n(x) \geq (1 - \alpha)\phi(x)$, ie $x \in E_n$.

We write

$$\begin{aligned}\lim_{n \rightarrow \infty} \int_E f_n &\geq \lim_{n \rightarrow \infty} \int_{E_n} f_n = \lim_{n \rightarrow \infty} \int_{E_n} (1 - \alpha)\phi \\ &= (1 - \alpha) \lim_{n \rightarrow \infty} \int_{E_n} \phi = (1 - \alpha) \int_E \phi.\end{aligned}$$

(we used that the E_n 's are increasing). Since this holds for all ϕ , we have $\lim_{n \rightarrow \infty} \int_E f_n \geq (1 - \alpha) \int_E f$. Let $M \in [0, \infty]$. Define $F_n = f_n^{-1}([M, \infty]) \cap F$, so the F_n 's are increasing. We have $\bigcup_{n \in \mathbb{Z}^+} F_n = F$, so

$$\lim_{n \rightarrow \infty} \int_F f_n \geq \lim_{n \rightarrow \infty} \int_{F_n} f_n \geq \lim_{n \rightarrow \infty} \int_{F_n} M = \lim_{n \rightarrow \infty} M\mu(F_n) = M\mu(F)$$

Since M is arbitrary, we have $\lim_{n \rightarrow \infty} \int_F f_n \geq \infty \cdot \mu(F) = \int_F f = (1 - \alpha) \int_F f$. Thus, $\lim_{n \rightarrow \infty} \int_X f_n \geq (1 - \alpha) \int_X f$. This holds for all α , so $\lim_{n \rightarrow \infty} \int f_n \geq \int f$. The reverse direction is clear. *****wrong